10236 - Resolve AGN Quenching in Our Backyard: An Infrared Investigation of Nearby Post-starburst Nuclei

Cycle: 5, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

The link between AGN and the quench of star formation (SF) in their host galaxies is a key question in galaxy evolution. While simulations and observations suggest strong AGN feedback expelling or heating the gas, AGN in nearby recently and rapidly quenched galaxies are often weak and/or heavily obscured. Resolving whether and how common low-luminosity AGN affect the gas to quench SF requires studies in spatially resolved regions of interest. Nearby post-starburst galaxies (PSBs) that have recently and rapidly quenched in the last 1 Gyr are ideal laboratories to study the interplay between AGN and gas since the onset of quenching. We propose a uniform analysis of the three nearest PSBs at sub-100 pc scales with archival JWST MIRI MRS data. We will measure outflow properties from asymmetric line profiles in both ionized and molecular emission lines, construct maps of star formation rate, fractional contribution of AGN to total MIR luminosity, H2 temperature and mass. By examining spatial correlations between regions of suppressed SF, AGN, and hot/turbulent gas, we will directly assess of whether quenching is associated with gas heating and/or disruption, and whether AGN can be the primary driver. This proposal, while limited in sample size, fully leverages all available archival observations on nearby PSBs and will provide the first detailed spatially resolved view of the interplay between quenching mechanisms. Our results will not only provide valuable details on the physical conditions under which rapid quenching happens, but also offer broader insights into out-of-field topics such as PAH variations, physics of AGN feedback, and rapid quenching at high redshifts.

OBSERVING DESCRIPTION

We propose to observe two low-redshift, dusty post-starburst galaxies using MIRI MRS with all four channels and three gratings to achieve the full coverage of 5–28 microns. This observing strategy will cover crucial diagnostic features and match the archival data we will incorporate in our analysis. With a spectral extraction radius of 0.4”, we require a continuum S/N >~3 in channel 2 medium and long gratings measured at the line-free regions at the bottom of the 9.7 micron Si absorption feature. Since the Si absorption is one of the faintest features in the MIR spectrum, this SNR will ensure spectra of higher quality in other channels, allowing us to successfully model the continuum and detect emission lines. We will take MRS background observations as non-interruptible sequence to the science observations, with simultaneous MIRI Imager observations on the galaxies in bands F770W, F1000W, F1130W to target PAH features at 7.7, 11.3 micron, and the Si absorption feature. For all observations, we use the 4-point dither pattern and the FASTR1 readout mode to optimize the dynamic range and minimize noise in a background-dominated regime. We do not require target acquisition exposures as our targets are extended and the blind JWST pointing accuracy is sufficient for our science goals.

Our GO-AR proposal will utilize a total sample of five post-starburst galaxies for analysis, including three from other programs: NGC 4418, IC 860 (PID 1991, data currently public) and NGC 1266 (PID 4972, data won’t be public by the start of this cycle; we will include it in analysis but not in budgets).